

Hack2Skill Ideathon Submission: Personal Gemini Journal

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Project ID: tensile-imprint-507111-j0

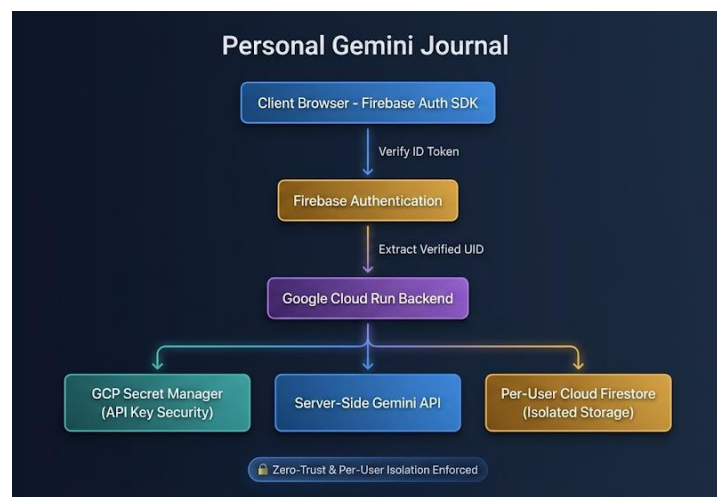
Live Endpoint (Cloud Run): <https://tensile-imprint-507111-j0.asia-southeast1.run.app>

Executive Summary

Personal Gemini Journal is an enterprise-grade, privacy-first workspace designed for intelligent journaling, reflective goal setting, and strategic brainstorming. Developed as part of the Gen AI Academy APAC Edition Ideathon, the application leverages custom security directives in Google AI Studio, a containerized Cloud Run backend, server-side Gemini API execution via Secret Manager, and per-user database isolation in Cloud Firestore.

Security & Architecture Overview

The system architecture prioritizes zero-trust principles and robust multi-tenant data isolation:

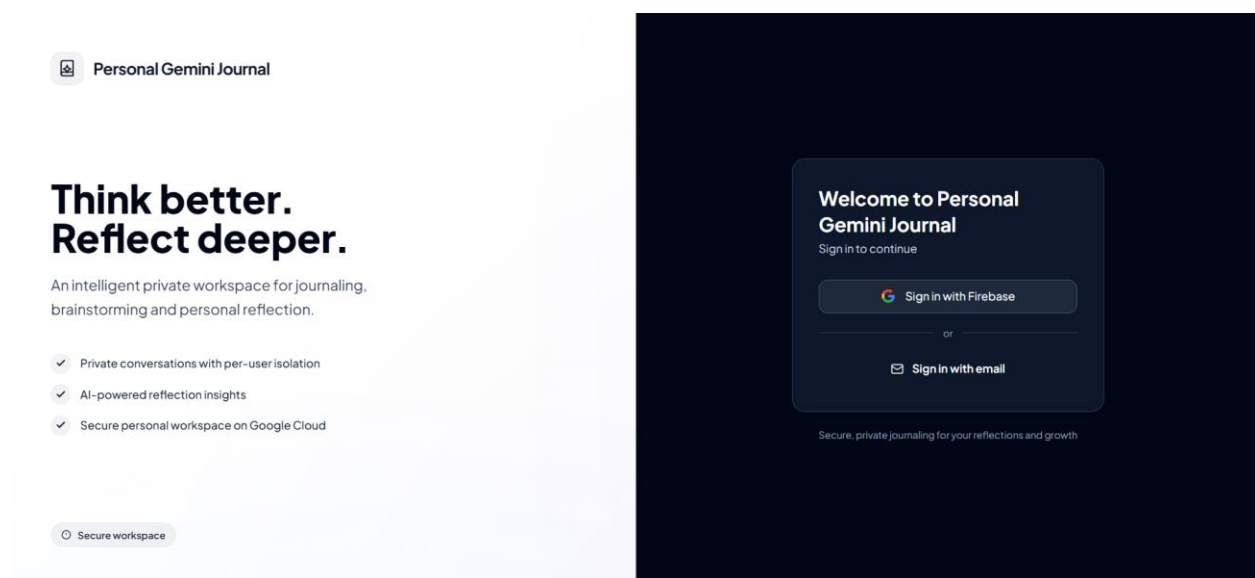


Core Security Pillars:

- **Authentication:** User identity is verified on every session using Firebase Authentication.
- **Authorization:** Backend endpoints extract and verify the `UID` directly from the session token rather than trusting front-end client payloads.
- **Per-User Isolation:** Cloud Firestore paths and rules restrict access so each user can only read and write their own documents
- **Secret Management:** The Gemini API key is fetched strictly on the server side via
- **Google Cloud Secret Manager** with least-privilege IAM permissions and is never exposed to the client browser.

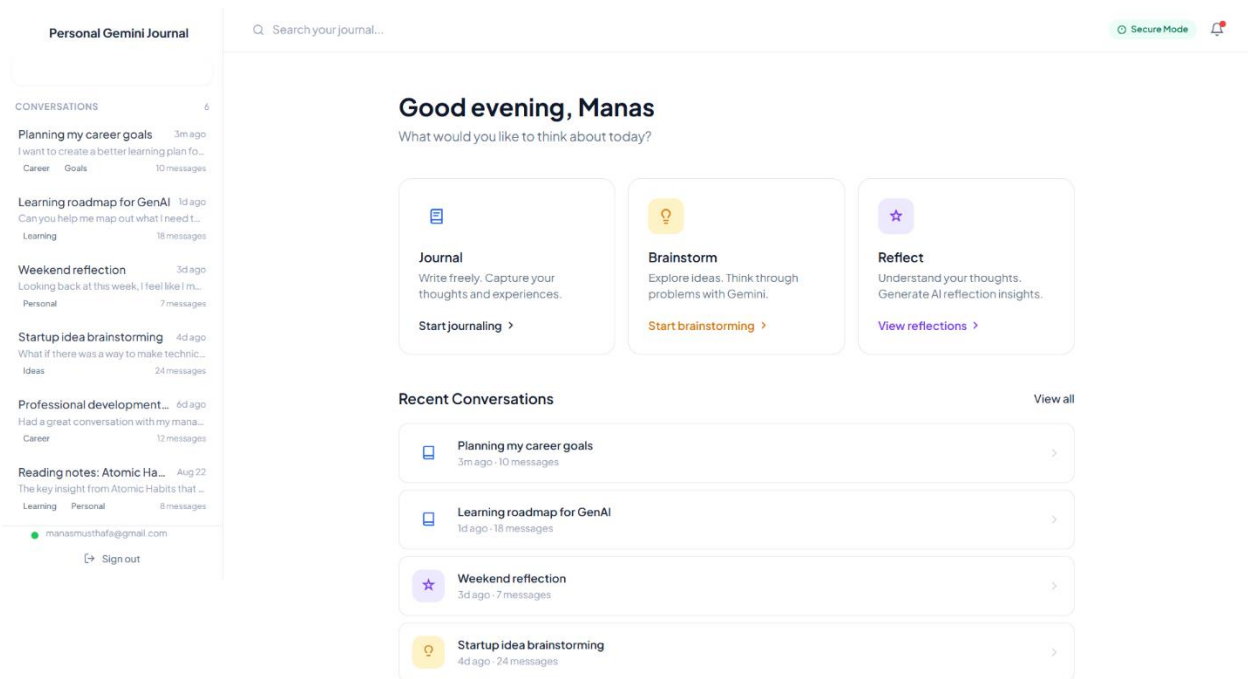
Detailed Application Screen Guide

Screen 1: Authentication Gateway



- **Overview:** Clean, secure landing page offering Single Sign-On (SSO) via Firebase Authentication and email/password sign-in.
- **Key Functionality:** Establishes the user identity boundary. Access to journal entries, AI generation, and database interactions is completely blocked until authentication succeeds.
- **Value Add:** Prevents unauthorized API usage and ensures all dynamic requests map directly to a verified user session.

Screen 2: Central Application Dashboard



- **Overview:** The primary command center displaying recent conversations, conversation tagging, thread counts, and fast actions (**Journal**, **Brainstorm**, **Reflect**).
- **Key Functionality:** Allows users to initiate targeted multi-turn conversations with Gemini or review previous entries categorized by tags such as *Career*, *Goals*, *Learning*, *Personal*, and *Ideas*.
- **Value Add:** Features real-time session loading from Firestore, fetching only the records belonging to the authenticated user's scope.

Screen 3: AI-Generated Reflection Engine

Personal Gemini Journal

CONVERSATIONS6

Planning my career goals7m ago

I want to create a better learning plan fo...

CareerGoals10 messages

Learning roadmap for GenAI1d ago

Can you help me map out what I need L...

Learning18 messages

Weekend reflection3d ago

Looking back at this week, I feel like I m...

Personal7 messages

Startup idea brainstorming4d ago

What if there was a way to make technic...

Ideas24 messages

Professional development...6d ago

Had a great conversation with my mana...

Career12 messages

Reading notes: Atomic Ha...Aug 22

The key insight from Atomic Habits that ...

LearningPersonal8 messages

manasmustafa@gmail.com

Sign out

Q Search your journal...

Secure Mode

☆Reflections

AI-generated insights from your journal conversations

☆AI-Generated Reflection · 1h ago

Demo

Planning my career goals

Career growthContinuous learningBuilding practical experience

EMOTIONAL TONE

Thoughtful and motivated

POSITIVE OBSERVATIONS

You consistently focused on turning learning into practical outcomes, not just accumulating knowledge.

There's a clear preference for understanding "why" over memorizing "how."

You naturally gravitate toward building things that solve real problems.

AREAS TO REFLECT ON

You may benefit from setting smaller weekly milestones to maintain momentum during the foundational learning phase.

Consider whether your interest in agents is driven by genuine curiosity or by industry hype — both are valid, but they lead to different paths.

SUGGESTED NEXT STEPS

1. Define a specific six-month goal with a measurable outcome

2. Break it into monthly milestones with concrete deliverables

3. Set a weekly review cadence to track progress and adjust

REFLECTION QUESTION

What would meaningful progress look like three months from now — not in terms of what you'd know, but what you'd be able to do?

☆AI-Generated Reflection · 3d ago

Demo

Weekend reflection

Self-awarenessTime managementProgress vs. perfection

EMOTIONAL TONE

Honest and slightly self-critical

POSITIVE OBSERVATIONS

You're developing a stronger internal metric for "productive" vs "busy" — this is a valuable skill.

The self-critical tone suggests high standards, but be careful it doesn't become a demotivator.

You identified the feeling of wasted time without external prompting, showing genuine self-awareness.

AREAS TO REFLECT ON

Explore whether "wasted time" was actually rest or recovery that your mind needed.

Consider tracking your energy levels alongside time use — you might find that some "unproductive" hours were necessary.

SUGGESTED NEXT STEPS

1. Try a simple time audit for one week — just 15-minute intervals, no judgment

2. Identify your top 3 time-wasting patterns and design one countermeasure for each

3. Celebrate the progress you did make before critiquing what you didn't

REFLECTION QUESTION

If a good friend told you they had the same week you did, would you call it wasted?

☆AI-Generated Reflection · 4d ago

Demo

Startup idea brainstorming

InnovationProblem-solvingProduct thinking

EMOTIONAL TONE

Excited and exploratory

POSITIVE OBSERVATIONS

Your brainstorming naturally gravitates toward user-centric solutions, not just technical cleverness.

You generated three distinct directions quickly, showing strong divergent thinking.

The initial problem statement ("make docs actually useful") is specific enough to be actionable.

AREAS TO REFLECT ON

You might be jumping to solutions before fully mapping the problem space — who exactly finds docs useless and in what context?

Consider whether this is a "nice to have" improvement or a "must fix" pain point for your target users.

SUGGESTED NEXT STEPS

1. Talk to 5 developers about their documentation frustrations before building anything

2. Pick one direction and sketch a minimal viable experiment

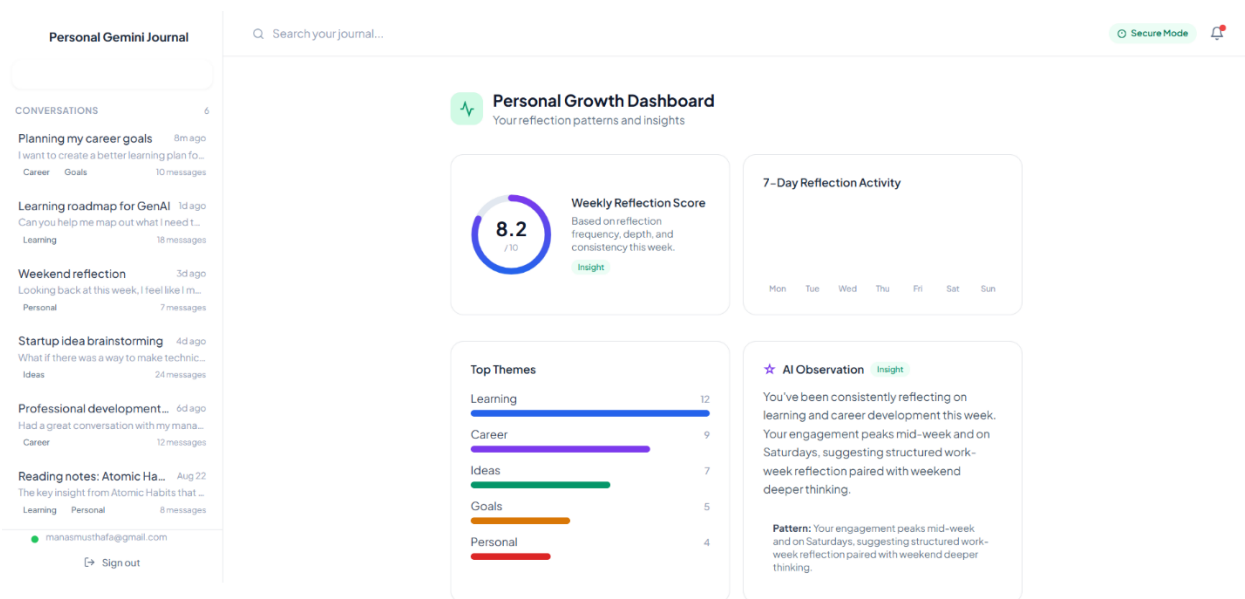
3. Write a one-paragraph "future press release" describing the solution as if it already exists

REFLECTION QUESTION

What would make you personally pay for better documentation — and would that align with what a company would pay for?

- **Overview:** A specialized metacognitive interface that processes journal threads into structured reflection outputs.
- **Key Functionality:** Automatically extracts key components from raw conversation history:
 - **Emotional Tone:** Synthesizes the user's underlying sentiment (e.g., *Thoughtful and motivated, Honest and slightly self-critical*).
 - **Positive Observations & Areas to Reflect On:** Identifies healthy habits and constructive improvement areas.
 - **Suggested Next Steps:** Generates structured, actionable milestones.
 - **Reflection Question:** Provides a tailored, open-ended question to encourage deeper thinking.
- **Value Add:** Built with Google AI Studio custom instructions to ensure insightful, empathetic, and production-safe formatting.

Screen 4: Personal Growth Dashboard (*Original Enhancement*)

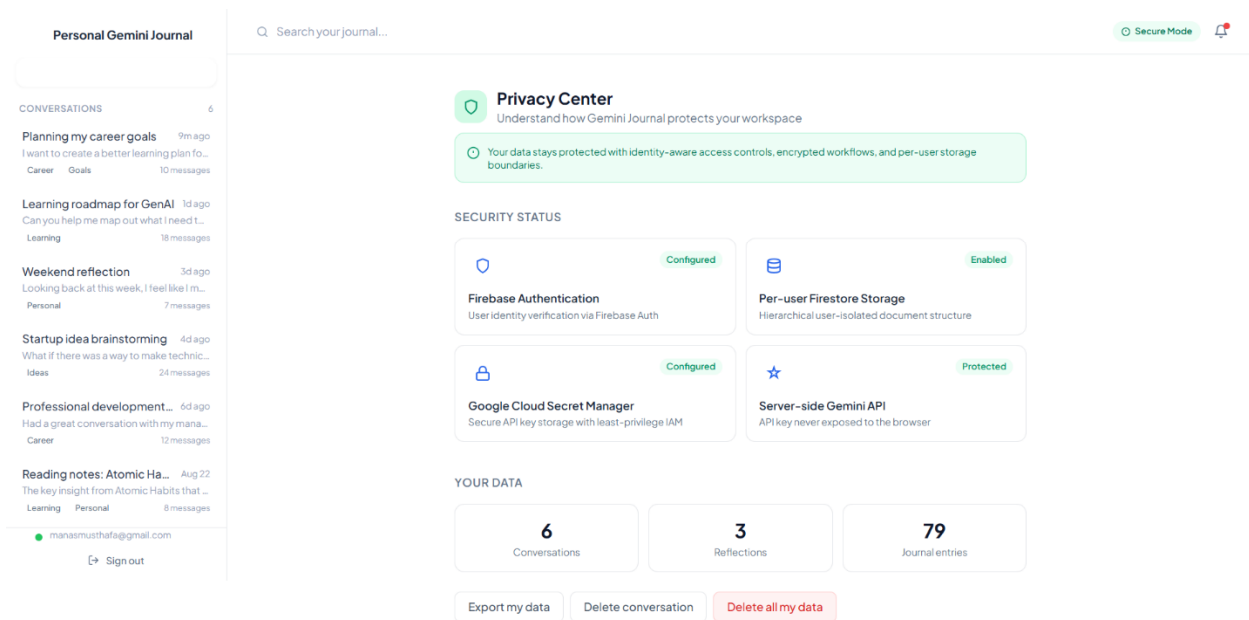


- **Overview:** An analytical view tracking user reflection patterns and consistency over time.
- **Key Functionality:** Displays quantitative and qualitative metrics including a **Weekly Reflection Score** (e.g., 8.2/10), a **7-Day Reflection Activity** chart,

a distribution of **Top Themes** (*Learning, Career, Ideas, Goals*), and synthesized **AI Observations** regarding thinking habits.

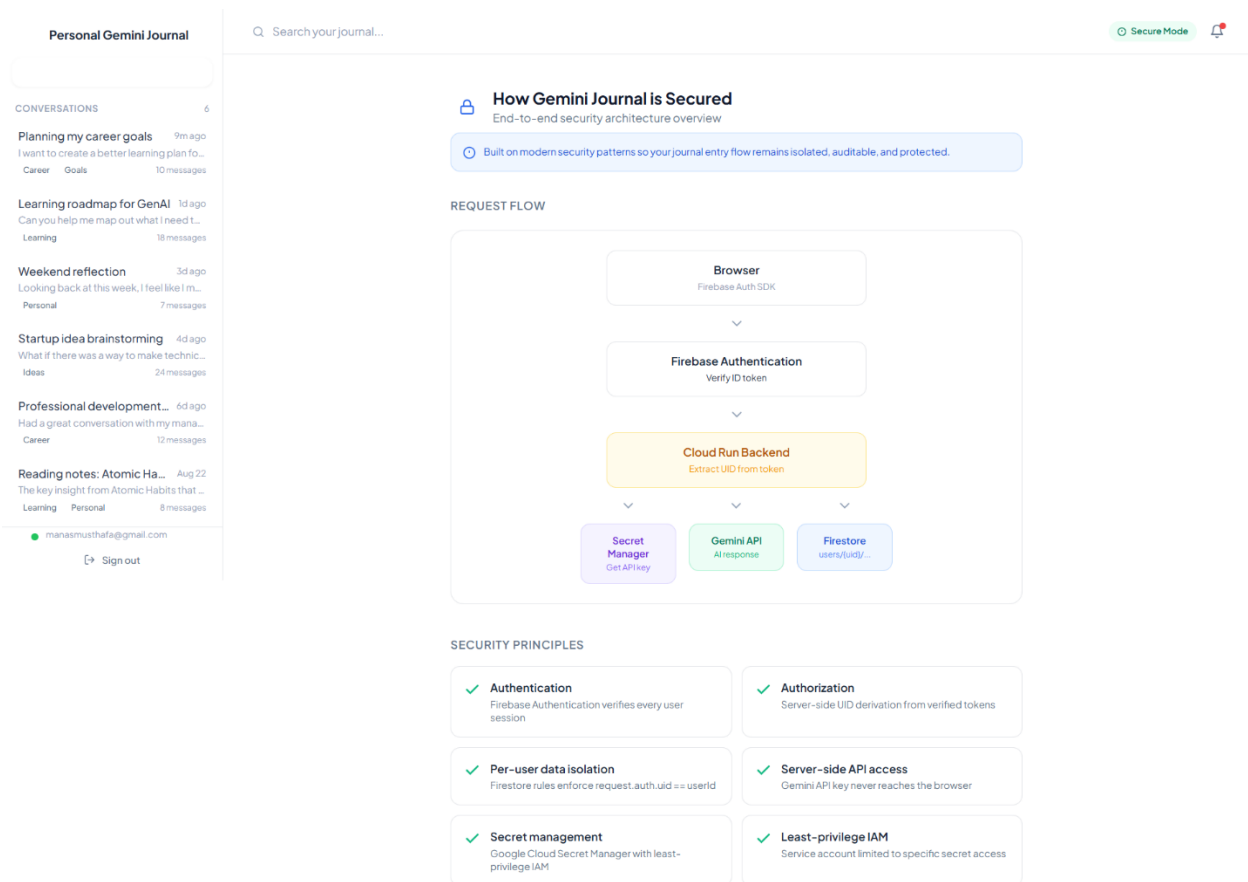
- **Value Add:** Goes beyond basic chatbot workflows by converting unstructured text logs into actionable productivity and self-growth metrics.

Screen 5: Privacy Center & Data Control



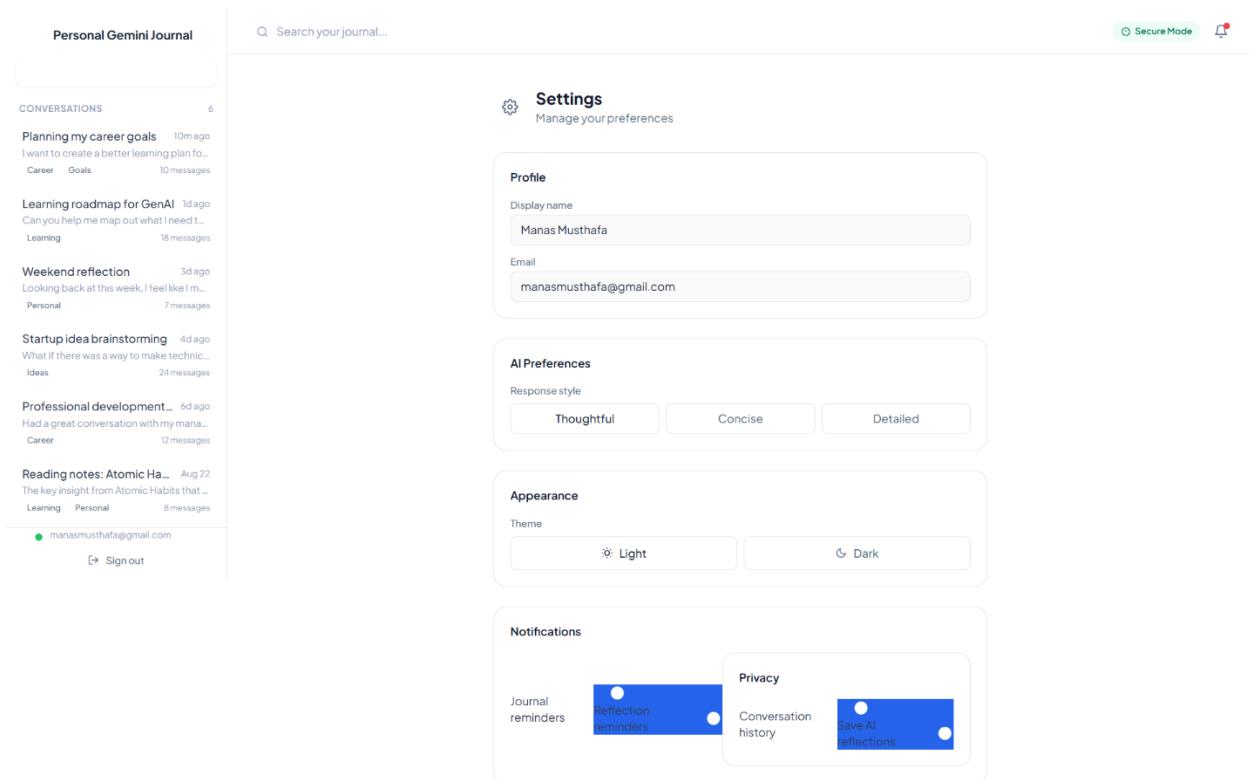
- **Overview:** A user-facing privacy management page showing live security statuses and data management controls.
- **Key Functionality:** Displays active safeguards (Firestore Auth, Isolated Firestore, Secret Manager, Server-side API execution) alongside total account usage stats (Conversations, Reflections, Journal Entries).
- **Value Add:** Empowers users with full data sovereignty through one-click **Export My Data**, **Delete Conversation**, and **Delete All My Data** actions.

Screen 6: System Security Architecture Diagnostic



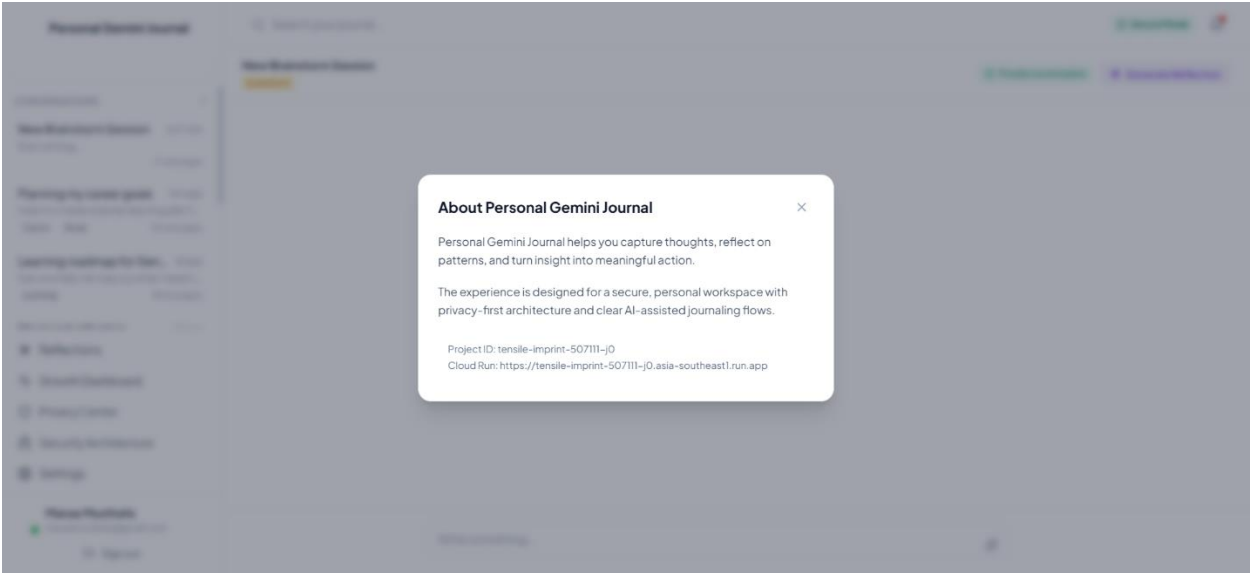
- **Overview:** An in-app architecture blueprint illustrating request lifecycle enforcement.
- **Key Functionality:** Details how requests pass through Firebase Auth ID token verification, server-side UID extraction on Cloud Run, Secret Manager retrieval, and isolated database queries.
- **Value Add:** Demonstrates total adherence to the Ideathon's security requirements directly within the product.

Screen 7: User Preferences & Customization



- **Overview:** User settings panel for managing user profile details, AI interaction styles, and theme settings.
- **Key Functionality:** Allows users to choose their preferred AI **Response Style** (*Thoughtful, Concise, Detailed*), toggle visual themes (*Light / Dark*), and configure privacy toggles like saving conversation histories or reflections.
- **Value Add:** Directly adjusts prompt constraints on the backend to customize the Gemini API behavior per user.

Screen 8: Identity & Deployment Verification Modal



- **Overview:** Modal detailing application metadata and cloud runtime environment status.
- **Key Functionality:** Confirms the active Google Cloud Project ID (tensile-imprint-507111-j0) and the regional Cloud Run URL endpoint.
- **Value Add:** Provides evaluation judges with immediate confirmation of live, verified deployment details.

Submission Criteria Checklist Alignment

Evaluation Criteria		Implementation Proof
 Authenticity		Designed and shipped original features beyond the basic lab, including the Personal Growth Dashboard, Privacy Center, and custom AI prompt tuning
 Usability		Single Sign-On integration via Firebase, intuitive navigation across journal modes, responsive layout, and dark mode support
 Stability		Containerized microservice deployed to Cloud Run with full server-side exception handling
 Security		Zero API key exposure on the frontend via GCP Secret Manager, strict server-side UID token extraction, and Firestore per-user document isolation